

ADVANCED DATA ANALYSIS FOR SUPERSTORE DATASET

Introduction

The advanced business analysis was conducted using the **Sample Superstore dataset**, which contains sales transaction data from **2014 to 2017**. The dataset includes information on orders, customers, products, categories, sub-categories, regions, customer segments, sales, quantity, discounts, and profit.

The purpose of the analysis was to identify **sales trends, profitability issues, product and regional performance, customer behaviour, seasonal patterns, and the effect of discounting on profitability**. Microsoft Power BI and DAX measures were used to create interactive dashboards, charts, tables, and slicers for analysis by year, region, category, and customer segment.

Dataset Overview

The dataset contains **9,994 transaction records, 5,009 distinct orders, and 793 distinct customers** covering four years. Key variables include Order ID, Order Date, Customer ID, Customer Name, Segment, Region, City, Category, Sub-Category, Product Name, Sales, Quantity, Discount, and Profit.

These variables provide sufficient information for analysing **sales, profitability, products, customers, geographical performance, discounts, and trends over time**.

Overall Sales and Profitability Performance

The business generated approximately **\$2.30 million in sales** and **\$286,397 in profit**, giving an overall **profit margin of 12.47%**. Although the business is profitable overall, profitability differs significantly across products, categories, regions, customers, and discount levels.

The dataset contains **5,009 distinct orders**, with an average sales value of approximately **\$458.61 per order**. There are **793 customers**, while the average discount is approximately **15.62%**. These KPIs provide the foundation for the detailed analysis.

Sales Performance Analysis

1. Yearly Sales Performance

Sales declined slightly from approximately **\$484,247 in 2014 to \$470,533 in 2015**, before increasing strongly to **\$609,206 in 2016** and **\$733,215 in 2017**.

Sales growth was **-2.83% in 2015, 29.47% in 2016, and 20.36% in 2017**. This shows a decline in 2015 followed by strong recovery and sustained growth in 2016 and 2017. Profit also increased from approximately **\$49,544 in 2014 to \$93,439 in 2017**, indicating improved financial performance.

2. Quarterly Performance

Quarterly analysis shows that **Q4 generally records the strongest sales performance**. In 2017, Q4 sales reached approximately **\$280,054**, compared with **\$123,145 in Q1**.

The stronger fourth-quarter performance indicates increased demand toward the end of the year. The business should therefore prepare **inventory, staffing, and marketing campaigns** before peak periods.

3. Monthly Performance

Sales vary considerably across months. The strongest months include **November (\$352,461), December (\$325,294), September (\$307,650), and March (\$205,005)**. **February** records the lowest sales at approximately **\$59,751**.

The results demonstrate clear seasonal variation. Inventory, staffing, and promotional planning should therefore take high- and low-demand periods into consideration.

Profitability Analysis

Total profit is approximately **\$286,397**, with an overall margin of **12.47%**. However, profitability differs substantially across categories and sub-categories.

1. Profit by Category

Technology is the strongest category, generating approximately **\$836,154 in sales and \$145,455 in profit**.

Office Supplies generates approximately **\$719,047 in sales and \$122,491 in profit**.

Furniture generates approximately **\$742,000 in sales but only \$18,451 in profit**. This makes Furniture a major profitability concern because its sales are high, but its contribution to profit is much lower than Technology and Office Supplies.

2. Profit by Sub-Category

The major loss-making sub-categories are **Tables, Bookcases, and Supplies**. Tables records approximately **-\$17,725 in profit**, Bookcases approximately **-\$3,473**, and Supplies approximately **-\$1,189**.

Tables is therefore the largest loss-making sub-category. This confirms that high sales or sales volume do not necessarily result in strong profitability.

Product Performance Analysis

The analysis identifies both highly profitable and loss-making products.

The **Canon imageCLASS 2200 Advanced Copier** is the strongest product by profit, generating approximately **\$25,200**. Other strong products include the **Fellowes PB500 Electric Punch Plastic Comb Binding Machine** and **Hewlett Packard LaserJet 3310 Copier**.

In contrast, the **Cubify CubeX 3D Printer Double Head Print** generates approximately **-\$8,880 in profit**, making it one of the largest individual product losses. Other loss-making products include the **Lexmark MX611dhe Monochrome Laser Printer** and **Cubify CubeX 3D Printer Triple Head Print**.

The results show that products should be evaluated using **sales, profit, and profit margin**, because high revenue does not always mean that a product is financially attractive.

1. High Sales but Low Profit

Some products generate significant sales but relatively low or negative profit. This is an important business risk because focusing only on revenue can make an underperforming product appear successful.

Management should therefore consider **both sales and profitability** when making product, pricing, and promotional decisions.

Regional Performance Analysis

Regional performance differs considerably. The **West region** is the strongest performer, generating approximately **\$725,458 in sales and \$108,418 in profit**. The **East** generates approximately **\$678,781 in sales and \$91,523 in profit**.

The **Central region** generates approximately **\$501,240 in sales and \$39,706 in profit**, while the **South region** generates approximately **\$391,722 in sales and \$46,749 in profit**.

West provides a useful benchmark for the other regions. **Central** requires further investigation because its sales are considerably higher than South's, but its profit is relatively low. Regional growth analysis also indicates that all regions have experienced growth, although growth rates differ.

Customer Segment Analysis

The three customer segments contribute differently to business performance.

The **Consumer segment** is the strongest, generating approximately **\$1.16 million in sales and \$134,119 in profit**.

Corporate generates approximately **\$706,146 in sales and \$91,979 in profit**, while **Home Office** generates approximately **\$429,653 in sales and \$60,299 in profit**.

Consumer customers therefore contribute the largest share of sales and profit. The business should maintain strong relationships with Consumer customers while identifying opportunities to increase the contribution of Corporate and Home Office customers.

Customer Performance

Customer analysis shows that high sales do not necessarily mean high profitability.

Sean Miller generates approximately **\$25,043 in sales** but produces a loss of approximately - **\$1,981**. In contrast, **Tamara Chand** generates approximately **\$19,052 in sales and \$8,981 in profit**, making the customer significantly more profitable despite lower sales.

Therefore, customer value should be assessed using **both revenue and profit**, rather than sales alone.

Discount Analysis

Discount analysis is one of the most important findings. The average discount is approximately **15.62%**, and profitability decreases as discount levels increase.

At **0% discount**, the profit margin is approximately **29.51%**, while at **20% discount**, it falls to approximately **11.82%**. At **30% discount**, the business records approximately **-\$10,369 in profit**. Losses increase to approximately **-\$23,057 at 40%**, **-\$20,506 at 50%**, and **-\$40,075 at 70%**.

The results indicate that discounts of **30% and above are associated with negative profitability**, making excessive discounting a major business risk.

Management should review high discount levels and determine whether the additional sales generated justify the reduction in profit. Discounts should be used strategically to increase sales without significantly damaging profit margins.

Business Problems and Opportunities

The analysis identified four major business problems and opportunities.

High Sales but Low Profit: Furniture generates approximately **\$742,000 in sales but only \$18,451 in profit**. This requires further investigation into its costs, pricing, and discount levels.

Loss-Making Products: Several products generate significant losses, particularly the **Cubify CubeX 3D Printer Double Head Print**. These products should be reviewed to determine whether their prices, costs, or discount levels should be changed.

Excessive Discounting: Discounts of **30% and above** are associated with negative profitability. The business should therefore introduce better discount controls and evaluate the profitability of promotional activities.

Regional Profitability Differences: The **West** is the strongest region, while **Central** produces relatively low profit compared with its sales. Weaker regions should be investigated to identify opportunities for improving profitability.

Conclusion

The advanced analysis shows that the Superstore has **strong overall sales and positive profitability**, but performance varies considerably across products, categories, regions, customers, and discount levels.

The main findings are that **Technology is the strongest category**, **Consumer is the largest customer segment**, **West is the strongest region**, and **sales experienced strong growth after 2015**. However, Furniture, Tables, and several individual products generate low or negative profit. The analysis also shows that high discount levels are strongly associated with declining profitability.

Management should therefore focus not only on increasing sales but also on **improving profit margins, controlling excessive discounts, reviewing loss-making products, planning for seasonal**

demand, and improving weaker regions and customer accounts. These actions can help the business achieve more sustainable and profitable growth.